1861–64

The Yosemite Grant, signed by President Abraham Lincoln on June 30, 1864, preserved the natural grandeur of the Yosemite region of California for the public.

Red blood cells get their red color from the protein hemoglobin. Felix Hoppe-Seyler discovered its vital role in oxygen transport in 1864.

BRITISH PHYSICIST JAMES CLERK MAXWELL REVEALED the pivotal advances he had made in the science of electricity and magnetism in two books: *On the Lines of Physical Force* (1861) and *A Dynamical Theory of the Electromagnetic Field* (1864). He began by explaining how an **electromagnetic field is created by waves** that radiate outward. He proved that these waves radiate at exactly the same speed as light, which showed that light is also an electromagnetic wave. Finally, he summed up his findings in four equations, now known as **Maxwell's equations**, which underpin all calculations relating to electricity and magnetism, in the same way that Isaac Newton's equations underpin all studies of motion.

In 1861, Maxwell also took the **first color photograph**. He had already proved that we see colors as varying intensities of three colors (see panel, right), but to demonstrate this he asked photographer Thomas Sutton (1819–75) to take three black and white photographs of a tartan ribbon, each one through a different color filter—red, green, and blue. He then projected all three images together, through filters of the same colors. The three color images mixed to recreate the picture in full color.

The colors found in the spectrum of sunlight by the Swedish physicist **Anders Ångström** (1814–74) in 1861, showed that the **Sun contains hydrogen gas**.

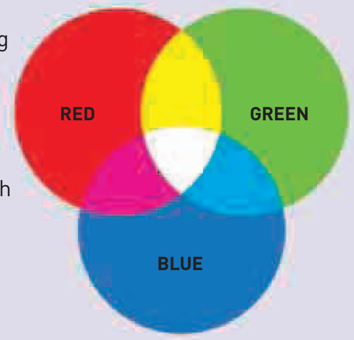
In the same year, French physician Paul Broca (1824–80) discovered a **key area of the brain for speech** in a man who had suffered a brain injury that rendered him unable to talk but able to understand. When the man died Broca performed an autopsy that revealed damage to a region of the brain now known as **Broca's area**.

After fossils of the *Glossopteris* fern were found in Africa, India, and South America, Austrian geologist **Edward Suess** (1831–1914) theorized in 1861 that these three continents were once joined by land bridges to create

The Berlin *Archaeopteryx*
The Berlin fossil of Archaeopteryx*, found in 1874, is the most complete. It plainly shows the toothed, dinosaur-like beak and feathered, birdlike wings.*

THREE-COLOR SYSTEM

Mixing three colors of light—red, green, and blue—in varying proportions makes every color in the rainbow. Images can be reproduced in full color by registering and then displaying how much light there is of each of these three colors in each part of the picture. Color was created this way in everything from the first color photograph to modern phone displays.



> **“…LIGHT CONSISTS** IN THE TRANSVERSE UNDULATIONS OF THE **SAME MEDIUM** WHICH IS THE CAUSE OF **ELECTRIC AND MAGNETIC PHENOMENA.”**

James Clerk Maxwell, British physicist, January 1862

1861 James Clerk Maxwell advances the study of **electromagnetism**

1861 Anders Ångström discovers that the **sun contains hydrogen gas**

1861 Maxwell creates the **first color photograph**

1861 Paul Broca names **Broca's area**, a key part of the brain for speech

1861 The first almost complete fossil of ***Archaeopteryx*** is found

1861 Edward Suess suggests the past existence of a **southern supercontinent**

January 31, 1862 American astronomer Alvan Graham Clark discovers the **white dwarf star Sirius B**

1861 American inventor Richard J. Gatling invents the **Gatling gun**

1862 Louis Pasteur makes early studies of the **germ theory of disease**

1862 Maxwell shows **light must be an electromagnetic wave**

1863 French writer **Jules Verne** publishes his **first science fiction book**, *Five Weeks in a Balloon*

1863 British astronomer William Huggins shows that **stars are made of the same gases as the Sun**